Short-Term Orientation, Trust and AI-Assisted Investment Behavior

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The survey data and Stata do-file are available at
https://github.com/iaochenacademic-lab/Short-Term-Orientation-Trust-and-AI-Assisted-Investment-Behavior

Abstract

A growing number of investors now rely on AI tools rather than on human advisers. This study asks whether investors with a stronger short-term orientation rely more on such tools, and whether trust in the tools strengthens that link. Prospect theory, scarcity theory and time discounting suggest that financial pressure shortens investors' time horizons. The ability, benevolence and integrity (ABI) model of trust helps explain when that pressure turns into reliance on AI. The data come from a survey of 162 adults in Macau, collected through snowball sampling in the spring of 2026. Factor analysis separates three constructs, namely short-term orientation, trust in AI and AI-assisted investment decisions, but it cannot tell the three dimensions of trust apart. In regressions with demographic controls, a stronger short-term orientation goes with more use of AI tools, and trust has a positive effect of its own. Trust also strengthens the link between the two. This moderating effect is modest and only marginally significant in conventional tests. Its bootstrap confidence interval still excludes zero, and the effect of short-term orientation is about twice as large among high-trust respondents as among low-trust ones. The moderation shows up in reported use rather than in stated intention, which suggests that trust matters most when money is at stake. Because the survey is cross-sectional, these results are associations rather than causal effects. Platforms should show users whose interests a recommendation serves, and regulators should watch for short-term reliance on similar AI tools spreading across many investors at once.

Keywords: Short-term orientation, Investment decisions, Trust, Behavioral finance, AI-assisted investing

1. Introduction

Since the global outbreak of COVID-19 in 2020, supply chain disruptions, geopolitical conflicts and global inflation have pushed the world economy into a downward cycle. With income expectations unstable and traditional paths to wealth blocked, more investors have stopped relying only on intuition or human advisers and have turned to artificial intelligence (AI) for investment decisions. Betterment, a robo-adviser, reported more than one million customers and over US$65 billion in assets under management (Betterment, 2025). The trend has also attracted fraud. In 2024 the Office of Investor Education and Advocacy of the Securities and Exchange Commission (SEC), the North American Securities Administrators Association (NASAA) and the Financial Industry Regulatory Authority (FINRA) issued a joint alert. It warned investors about schemes that claim to use AI, including some sold as robo-advisers (Arkansas Securities Department, 2024).

Yao (2024) illustrated that AI is becoming an integral part of modern investing practices because AI empowers investors with immediate decision-making assistance without sacrificing speed and accuracy. Much less is known about the psychological side of this choice, especially in a downturn. Why do some investors hand decisions to an algorithm while others do not? This study looks at two factors. The first is short-term orientation, the tendency to weigh immediate outcomes more heavily than distant ones. The second is trust in the AI tool. The study examines whether investors with a stronger short-term orientation use AI tools more, and whether trust strengthens that relationship.

The study brings together two lines of research that are usually kept apart, work on time preference and work on technology adoption. It also treats trust as a condition that changes how strongly time preference is linked to AI use, not only as a direct predictor. This condition turns out to operate when investors act rather than when they state intentions. The evidence comes from Macau, a setting that has received little attention in this literature.

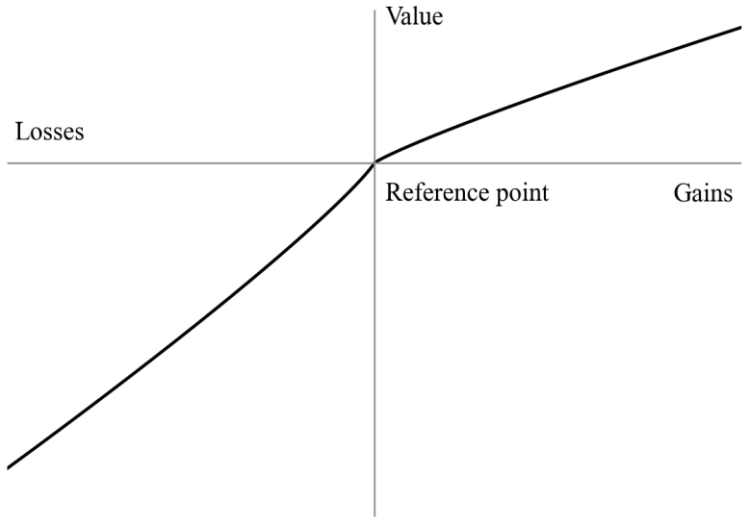
Section 2 reviews research on time preference and on trust in algorithms. Section 3 sets out the research model, the measures and the hypotheses. Section 4 reports the results. Section 5 concludes, and Section 6 discusses limitations and directions for future research.

2. Literature review

2.1 Short-term-oriented investment under economic stress

Existing studies situate this behavior within frameworks of motivation theory and cognitive psychology, demonstrating how economic strain alters individuals' temporal and risk preferences, thereby promoting reliance on short-term-oriented technologies. Prospect theory (Kahneman & Tversky, 1979) has long shown that people become risk seeking when they face certain losses, and Figure 1 shows the value function behind this prediction. After 2020, volatile markets left many investors below their reference point. The theory predicts that such investors take more risks in the hope of recovering losses quickly, for example through speculative short-term trading. Brain studies link loss aversion to the amygdala (De Martino et al., 2010), which suggests that this response is partly emotional. Seen this way, an AI tool that promises fast analysis may appeal to investors partly because it offers a way to act on that urge.

Figure 1. Value function in prospect theory

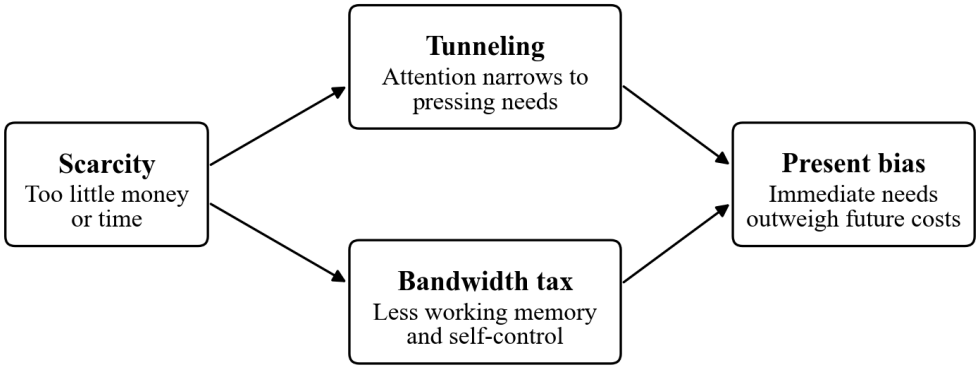


Note: The value function is defined over gains and losses relative to a reference point. It is concave for gains, convex for losses and steeper for losses than for gains. The curve is illustrative. Adapted from Kahneman and Tversky (1979).

Scarcity theory (Mullainathan & Shafir, 2013) adds a cognitive explanation. In this view, scarcity goes beyond a shortfall of resources and imposes a psychological bandwidth tax. The tax narrows attention to the most pressing needs, a process the authors call tunneling, uses up working memory and executive control, and makes complex decisions harder (Figure 2). Tunneling leads people to focus on short-term performance at the expense of longer-term issues. According to Yao (2024), AI automates data processing, screening and routine analysis, which takes over tasks that demand attention. AI also lowers search and evaluation costs and can apply time-saving heuristics, freeing limited bandwidth for higher-order judgments in the short term. Compensatory control theory points the same way. When the outside world feels chaotic and uncontrollable, people look for external systems that seem to impose order (Kay et al., 2008). A tool that turns noisy market data into clear, structured advice offers that kind of order. Investors who are short of attention may therefore treat AI as a cognitive aid that helps them keep a sense of control.

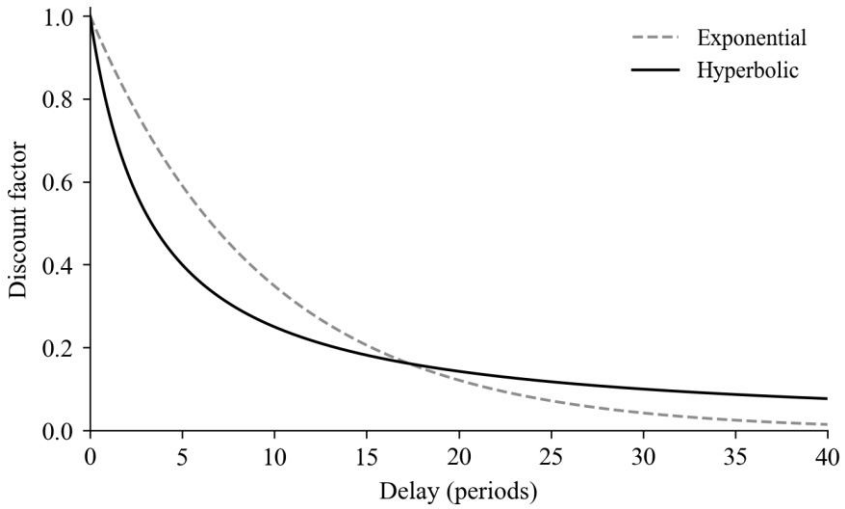
Beyond risk, economic pressure also changes an investor's sense of time. Research on intertemporal choice shows that people discount rewards more steeply over short delays than over long ones, a pattern known as hyperbolic discounting or present bias (Loewenstein & Prelec, 1992; Laibson, 1997). Figure 3 shows the steep early drop of the hyperbolic curve. Financial pressure is likely to make this bias stronger, and AI tools that respond within seconds suit a preference for quick feedback. Individual investors also tend to buy stocks that have just caught their attention, such as those in the news or with unusual trading volume (Barber & Odean, 2008). Their trades respond strongly to recent returns as well (Grinblatt & Keloharju, 2001). Both findings point to decisions driven by what has just happened rather than by long-term plans.

Figure 2. Scarcity mindset



Note: Scarcity narrows attention and taxes cognitive capacity, and both effects push choices toward the present. Based on Mullainathan and Shafir (2013).

Figure 3. Hyperbolic and exponential discounting



Note: The exponential curve discounts at a constant rate per period. The hyperbolic curve falls faster at short delays and more slowly at long ones. The parameter values are illustrative. Based on Laibson (1997).

Short-term orientation and fast, automated tools may therefore reinforce each other. In equity markets, algorithmic trading has already raised the speed of trading and changed how liquidity is supplied (Hendershott et al., 2011). The picture is not wholly positive. Barber and Odean (2000) found that the households that traded most earned the lowest net returns, mainly because of trading costs. Many machine-learning models are also hard to interpret (Burrell, 2016). When the logic behind a recommendation is hidden, it becomes easier to avoid responsibility for the outcome, and after a loss investors may blame the algorithm rather than their own judgment.

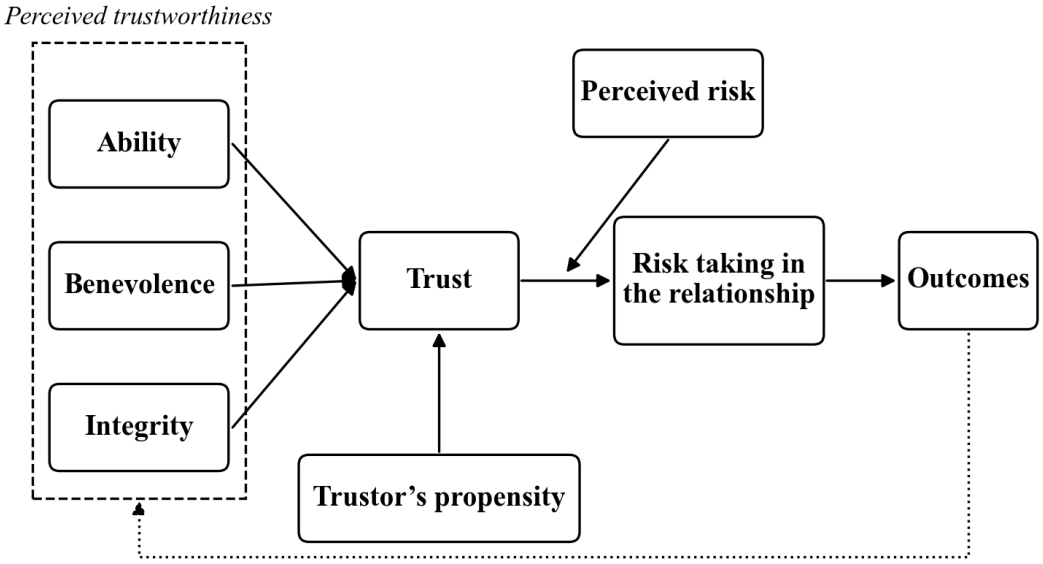
2.2 Trust transfer from people to AI

The reason people resort to AI can be explained through the lens of trust. Under Social Exchange Theory (Emerson, 1976), trust is a psychological contract built on interpersonal interaction and institutional guarantees, and it rests on expectations of predictability and reciprocity. The Theory of Planned Behavior (Ajzen, 1991) explains intentions through attitudes, social norms and perceived behavioral control. Studies of technology adoption often add trust as a belief that shapes these attitudes (Gefen et al., 2003). Trust also has a biological side. In one experiment, people given oxytocin were more willing to entrust money to others (Kosfeld et al., 2005).

In economic behavior, trust acts as a lubricant that reduces transaction costs and mitigates the fear of opportunistic behavior. High levels of trust facilitate capital mobility and long-term

commitment, allowing investors to delegate decision-making power to agents without constant monitoring. Mayer et al. (1995) argue that this trust rests on judgments of a party's ability, benevolence and integrity, the ABI model shown in Figure 4. In their model, trust leads to risk taking in the relationship, and perceived risk shapes how far it does so.

Figure 4. Integrative model of organizational trust



Note: Ability, benevolence and integrity shape perceived trustworthiness, which together with the trustor's propensity leads to trust. Trust leads to risk taking in the relationship, conditional on perceived risk, and outcomes feed back into perceived trustworthiness. Adapted from Mayer et al. (1995).

However, interpersonal trust can break down when there is a perceived lack of integrity or predictability. During downturns, human advisers may be blamed for forecasting failures, which is a question of ability, or suspected of conflicts of interest, which is a question of integrity. Low trust has real consequences. Guiso et al. (2008) show that people who trust less are less likely to hold stocks at all. In the 2008 global financial crisis, confidence in financial experts, bankers and government institutions fell sharply (Sapienza & Zingales, 2012).

When investors perceive human decision-makers as having uncontrollable biases, they may look for security in rule-based systems that seem more objective. In ABI terms, turning to AI means giving more weight to a system's computing ability and apparent neutrality. AI tools are also attractive because they handle large amounts of information quickly (Yao, 2024). From the perspective of the Technology Acceptance Model (TAM), this efficiency raises the perceived usefulness of AI. For investors whose cognitive bandwidth is exhausted, fast execution provides a sense of structure. How a system is presented also matters. Waytz et al. (2014) found that people trusted a simulated autonomous car more when it was given human-like features such as a name and a voice. Logg et al. (2019) found that people often prefer advice from an algorithm to the same advice from a person, a pattern they call algorithm appreciation.

This trust transfer is not entirely rational. A preference for algorithms can turn into over-reliance when users give a system more authority than its record justifies. Trust in algorithms is also fragile. In experiments by Dietvorst et al. (2015), people abandoned an algorithm that made a mistake more readily than a human forecaster who made the same mistake, a pattern they call algorithm aversion. Trust in AI can therefore be high before the first visible error and drop quickly afterwards. Burrell (2016) points out that the black-box nature of AI systems makes it hard for users to understand how decisions are made, which in investing makes risks harder to spot. Mayer et al. (1995) emphasize that trust should rest on ability, benevolence and integrity, yet many financial AI tools say little about how they reach their advice or whose interests it serves.

A last possibility concerns attribution. A loss that follows one's own decision can damage self-efficacy, whereas a loss that follows an algorithm's advice can be blamed on the tool. In a downturn, AI may therefore serve as a convenient target for blame. If responsibility shifts in this way, investors may also pay less attention to the risks they are taking.

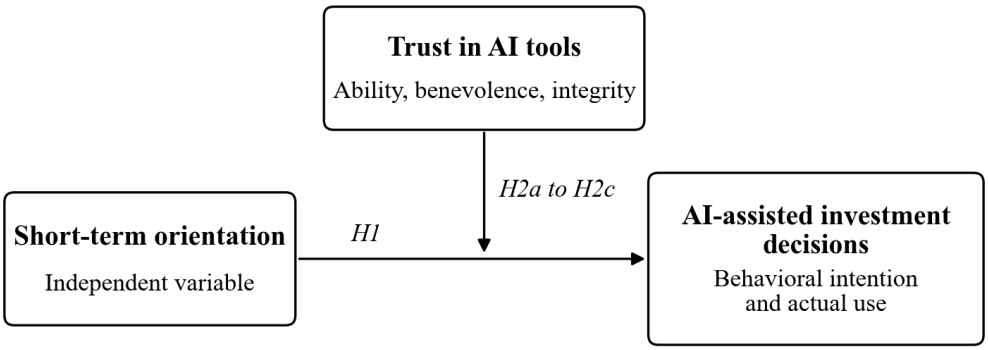
This review leaves two gaps. Research on time preference and research on trust in algorithms have rarely been brought together, so little is known about whether short-term-oriented investors are more drawn to AI tools. Trust, moreover, is usually treated as a direct predictor of adoption. It may also work as a condition, since an investor under pressure may want fast help from AI but act on that wish only if the tool seems trustworthy. This study tests that idea.

3. Research methodology

3.1 Research framework

Building on prospect theory, scarcity theory and time discounting, this study argues that economic stress can induce a short-term orientation among investors, making them favor quick decisions and immediate results. The literature suggests that such an orientation goes with a greater intention to use AI and with more actual use of it. AI then serves as a cognitive aid that restores a sense of order and control in a turbulent market. Figure 5 presents the research model. Short-term orientation is the independent variable and AI-assisted investment decisions are the dependent variable. Trust in the AI tool, with its dimensions of ability, benevolence and integrity, is the moderator that is expected to ease the transfer of reliance from people to technology. Economic stress itself is not measured in the survey. It is the background that motivates the model rather than a variable in it.

Figure 5. Research model



Note: H1 is the direct relationship between short-term orientation and AI-assisted investment decisions. H2a to H2c state that each dimension of trust strengthens this relationship.

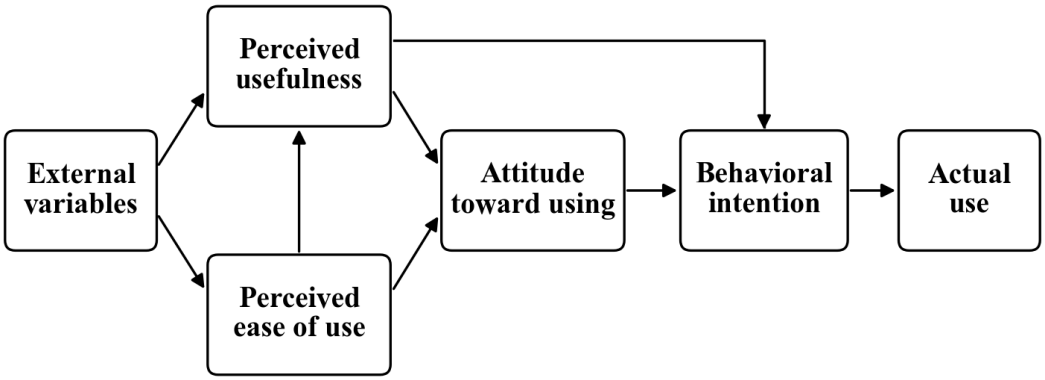
3.2 Variables

3.2.1 Dependent variable

This study uses investment decisions as the dependent variable. The Technology Acceptance Model (Davis, 1989) is the most widely applied model of how users accept and use new technology (Venkatesh, 2000). As Figure 6 shows, the model places behavioral intention between users' beliefs about a system and their actual use of it, and treats intention as the direct cause of use. Ajzen (1991) further pointed out in the Theory of Planned Behavior that intention and actual use are related but not equivalent, so the two need to be measured separately. This study therefore treats behavioral intention and actual use as the two dimensions of the dependent variable.

Behavioral intention (BI) is measured with three items adapted from the TAM scale of Davis (1989). Actual use (AS) is also measured with three items, which ask how often and how far participants relied on AI-assisted tools for actual investment decisions in the past three months. All six items use a five-point Likert scale, and their mean is the score of the dependent variable. A higher score means a stronger intention to use AI tools and more actual use of them. Panel A of Table 1 lists the items. Item AS3 asks whether the respondent would recommend the tool to others, which is closer to advocacy than to use, so Section 4.7 also reports results for a use score without it.

Figure 6. Technology Acceptance Model



Note: Adapted from Davis et al. (1989).

3.2.2 Independent variable

This study uses short-term orientation as the independent variable. As an expression of time preference, short-term orientation has been widely studied in behavioral economics. Loewenstein and Prelec (1992) show that the rate at which people discount future returns changes non-linearly with delay. Building on this idea, Strathman et al. (1994) developed the Consideration of Future Consequences (CFC) scale. It measures how much weight a person gives to immediate rather than distant consequences and has been validated in many studies.

Here the scale measures how strongly participants prefer immediate over long-term returns in their investment decisions. The seven items are adapted from the immediate subscale of the CFC and adjusted to the topic (Panel B of Table 1). As in Strathman et al. (1994), items are answered on a five-point scale, and a higher score means a stronger short-term orientation. Item I3 needs a comment. It is worded in the future-oriented direction and is reverse-scored in the CFC scale itself. In this sample, however, it correlates positively with the other six items, so respondents appear to have read it in the same direction as the rest. Reverse-scoring it would lower the internal consistency of the scale considerably. The item is therefore scored as worded, and Section 4.7 shows that the results hardly change when it is dropped.

3.2.3 Moderating variable

Trust is a core concept in research on technology adoption and human–automation interaction. The ABI model of Mayer et al. (1995) groups the sources of trust into three dimensions, which are ability, benevolence and integrity. Ability refers to the skills and knowledge needed to perform in a specific domain. Benevolence refers to actions driven by concern for the trustor's interests rather than one's own. Integrity refers to adherence to a set of principles, often related to impartiality. For automated systems specifically, Jian et al. (2000) developed a trust scale from an empirical study of how people describe trust in machines.

Following the ABI model, this study wrote twelve items, four each for ability (A), benevolence (B) and integrity (In), as listed in Panel C of Table 1. The ability items measure how far participants see the AI tool as professionally capable and reliable. The benevolence items measure whether they believe the tool puts their interests first, and the integrity items whether they believe it follows its stated principles. The mean of the twelve items is the overall trust score, with higher scores meaning stronger trust. All items use a five-point Likert scale. As Section 4.3 shows, respondents did not distinguish the three dimensions, so the overall score is used in the main model.

Before the interaction term is formed, both the trust score and short-term orientation are mean-centered (Aiken & West, 1991). The interaction term is the product of the two centered variables. Centering reduces multicollinearity, and the coefficient on the product measures the moderating effect of trust.

3.2.4 Control variables

The model also controls for age, gender, educational stage, monthly income and marital status, which are defined in Panel D of Table 1. Gender and marital status are categorical and enter as dummy variables. Ajzen's (1991) Theory of Planned Behavior holds that perceived behavioral control has a significant effect on intention, and here on investment decisions. Perceived behavioral control may also be related to short-term orientation. To reduce omitted-variable bias, this study controls for these demographic characteristics, which are related to perceived behavioral control.

Table 1. Variables and questionnaire items

Item	Statement or definition
Panel A. AI-assisted investment decisions (dependent variable, adapted from Davis, 1989)	
BI1	I am willing to use AI tools to assist my investment decisions.
BI2	I intend to continue using AI investment tools in the future.
BI3	When the market is uncertain, I prefer to refer to the analysis of AI tools rather than relying on my personal judgment.
AS1	I am currently using an AI tool to assist in investment decisions.
AS2	In the past three months, I have relied on this AI tool to make at least one actual investment decision.
AS3	I would recommend this AI investment tool to important people around me (family, friends, colleagues).
Panel B. Short-term orientation (independent variable, adapted from Strathman et al., 1994)	
I1	My behavior is only influenced by the immediate (i.e., a matter of days or weeks) outcomes of my actions.
I2	My convenience is a big factor in the decisions I make or the actions I take.
I3	I am willing to sacrifice my immediate happiness or well-being in order to achieve future outcomes.
I4	I generally ignore warnings about possible future problems because I think the problems will be resolved before they reach crisis level.
I5	I think that sacrificing now is usually unnecessary since future outcomes can be dealt with at a later time.
I6	I only act to satisfy immediate concerns, figuring that I will take care of future problems that may occur at a later date.
I7	Since my day to day work has specific outcomes, it is more important to me than behavior that has distant outcomes.
Panel C. Trust in AI tools (moderating variable, based on Mayer et al., 1995)	
A1	This AI investment tool has professional capabilities in analyzing financial data.
A2	This AI investment tool is reliable.
A3	This AI investment tool can perform analytical tasks with high precision.
A4	Even during market volatility, this AI investment tool still provides reliable advice.
B1	This AI investment tool will do everything in its power to help me.
B2	This AI investment tool truly cares about my needs.
B3	This AI investment tool prioritizes my financial interests over other considerations.
B4	This AI investment tool will not intentionally give advice that harms me.
In1	I never worry about whether this AI investment tool will follow its stated principles.
In2	This AI investment tool operates according to reasonable and consistent principles.
In3	This AI investment tool's decisions are impartial.
In4	Even without monitoring, this AI investment tool adheres to its stated operating principles.
Panel D. Control variables	
Age	Five age groups from 18 to 24 up to 55 and over, treated as continuous
Gender	Dummy variable equal to one for women
Education	Four levels from secondary or below to postgraduate, treated as continuous
Income	Five bands of monthly disposable income, treated as continuous
Marital status	Dummy variables for married and other, with unmarried respondents as the reference group

Note: All items in Panels A to C are rated on a five-point Likert scale, and Panels A and C were introduced with a hypothetical scenario. BI denotes behavioral intention and AS actual use. Items in Panel C measure ability (A), benevolence (B) and integrity (In). Each construct score is the mean of its items, so investment decisions average the six items in Panel A and overall trust averages the twelve items in Panel C.

3.3 Research hypotheses

Table 2 lists the hypotheses. They concern the direct effect of short-term orientation on AI-assisted investment decisions, covering both behavioral intention and actual use, and the moderating role of trust and its three dimensions. All hypotheses predict that short-term orientation, induced by economic stress and scarcity, increases reliance on AI tools. Trust is expected to strengthen this relationship. Higher perceived ability, benevolence and integrity should therefore tighten the link between short-term orientation and AI-assisted investment decisions.

Table 2. Research hypotheses

	Statement
H1	Short-term orientation is positively related to AI-assisted investment decisions.
H2a	The ability dimension of trust moderates the relationship between short-term orientation and AI-assisted investment decisions. The higher the perceived ability, the stronger the positive relationship.
H2b	The benevolence dimension of trust moderates the relationship between short-term orientation and AI-assisted investment decisions. The higher the perceived benevolence, the stronger the positive relationship.
H2c	The integrity dimension of trust moderates the relationship between short-term orientation and AI-assisted investment decisions. The higher the perceived integrity, the stronger the positive relationship.

3.4 Research method

This study adopted the questionnaire survey method, using a snowball sampling strategy to reach a broad range of participants. Data were collected both online through Google Forms and on site between late February and early April 2026, and 162 valid questionnaires were obtained. All participants were told the purpose of the study before taking part. Their answers were anonymous and kept confidential, in line with research ethics. The data were compiled in Microsoft Excel and analyzed in Stata 18.0. The analysis includes descriptive statistics, reliability tests, exploratory factor analysis, correlation analysis and moderated regressions, together with simple slopes, the Johnson–Neyman technique and bootstrapping.

3.5 Model

The analysis first regresses investment decisions on the control variables X alone, as in equation (1).

$$Invest_i = \beta_0 + \gamma X_i + \varepsilon_i \quad (1)$$

Short-term orientation and trust are then mean-centered, so that STO_i and $Trust_i$ denote deviations from their sample means. The benchmark model in equation (2) adds both variables and their product.

$$Invest_i = \beta_0 + \beta_1 STO_i + \beta_2 Trust_i + \beta_3 (STO_i \times Trust_i) + \gamma X_i + \varepsilon_i \quad (2)$$

Three further models replace *Trust* in equation (2) with ability, benevolence and integrity in turn. The coefficient β_3 measures moderation. Because both variables are centered, β_1 is the effect of short-term orientation at average trust, and its effect at any level of trust M equals $\beta_1 + \beta_3 M$. This conditional effect is evaluated at one standard deviation below and above mean trust (Aiken & West, 1991). The Johnson–Neyman technique (Johnson & Neyman, 1936) then identifies the range of trust over which it is significant at the 5% level. Because the sample is modest, the interaction coefficient is also bootstrapped with 2,000 resamples. All models are estimated by ordinary least squares (OLS).

4. Results

4.1 Descriptive statistics

Table 3 describes the sample. Men slightly outnumber women. Most respondents are between 25 and 44 years old, and nearly half fall in the group aged 35 to 44. About two in five report a monthly disposable income between MOP 25,001 and 35,000, and most of the rest earn

between MOP 15,001 and 25,000. The sample therefore consists mainly of middle-income workers in Macau. Nearly all respondents hold at least a bachelor's degree.

Table 3. Sample characteristics

Characteristic	Category	n	%
Gender	Male	87	53.70
	Female	75	46.30
Age	18 to 24	16	9.88
	25 to 34	60	37.04
	35 to 44	79	48.77
	45 to 54	7	4.32
	55 and over	0	0.00
Monthly disposable income (MOP)	15,000 or less	13	8.02
	15,001 to 25,000	51	31.48
	25,001 to 35,000	66	40.74
	35,001 to 45,000	28	17.28
	Over 45,000	4	2.47
Educational stage	Secondary or below	2	1.23
	Vocational secondary	7	4.32
	Undergraduate	105	64.81
	Postgraduate	48	29.63
Marital status	Unmarried	97	59.88
	Married	61	37.65
	Other	4	2.47

Note: N = 162. Percentages may not sum to 100 because of rounding.

Table 4 summarizes the scale scores. Each score is also split into low, medium and high groups, as defined in the note to the table. All scale means sit slightly below the midpoint of the five-point scale. For short-term orientation, the low group is the largest and the high group the smallest, so most respondents lean only moderately toward the short term. For the three trust dimensions, the low and medium groups are of similar size and the high group is the smallest. Respondents' trust in the competence, good intentions and honesty of AI tools is therefore moderate to low. Intention and actual use are more evenly spread, with the medium group the largest. About a quarter of respondents fall in the high group for actual use, so a sizable share have already brought AI into their investment decisions.

4.2 Reliability and correlation analysis

Cronbach's α is used to assess the internal consistency of each scale. Values above 0.8 are usually taken to indicate good reliability, and every scale in this study exceeds that level (Table 4). A high α does not by itself show that the scales measure different things, and Section 4.3 checks this.

Table 4. Scale scores and reliability

Scale	Items	α	Mean	SD	Low (%)	Medium (%)	High (%)
Short-term orientation	7	0.927	2.810	1.048	42.59	34.57	22.84
Trust in AI tools	12	0.954	2.825	0.980	35.80	41.98	22.22
Ability	4	0.860	2.847	1.031	37.65	37.04	25.31
Benevolence	4	0.881	2.799	1.016	38.27	38.89	22.84
Integrity	4	0.874	2.829	1.011	40.74	35.80	23.46
Investment decisions	6	0.911	2.877	1.012	29.01	45.68	25.31
Behavioral intention	3	0.824	2.889	1.084	27.16	43.21	29.63
Actual use	3	0.841	2.864	1.021	24.69	50.00	25.31

Note: N = 162. Scores are item means on a scale from 1 to 5. Low means a score of 2.33 or below, medium a score above 2.33 and up to 3.66, and high a score above 3.66. Scoring item I3 in reverse would reduce α for short-term orientation to 0.728.

Table 5 reports Pearson correlations among the main measures. All of them are positive and significant at the 1% level. The three trust dimensions are very strongly correlated with one another, so respondents who see an AI tool as capable also tend to see it as benevolent and honest. Behavioral intention and actual use are also very strongly correlated, which means that respondents who intend to use AI for investing usually report doing so. Short-term orientation is moderately correlated with both intention and use, and so is each dimension of trust. These patterns fit the expectation that trust comes before the adoption of AI-assisted investing, although correlations alone cannot establish the direction.

Table 5. Pearson correlations among the main measures

	(1)	(2)	(3)	(4)	(5)	(6)
(1) Short-term orientation	1					
(2) Ability	0.397***	1				
(3) Benevolence	0.418***	0.873***	1			
(4) Integrity	0.419***	0.889***	0.893***	1		
(5) Behavioral intention	0.409***	0.420***	0.419***	0.394***	1	
(6) Actual use	0.461***	0.376***	0.428***	0.430***	0.850***	1

Note: N = 162. *** p < 0.01, ** p < 0.05, * p < 0.1.

4.3 Construct validity

All scales were completed by the same respondents in one sitting, so it is worth checking whether they capture distinct constructs. An exploratory factor analysis of all 25 items, using principal-component factors with promax rotation, yields three components with eigenvalues above one. Together they explain more than two thirds of the variance. As Table 6 shows, the seven short-term orientation items load on one factor, the twelve trust items on a second and the six intention and use items on a third. Every item loads strongly on its own factor and weakly on the others.

Table 6. Factor structure and discriminant validity

Factor	Items	Loadings	Largest cross-loading
Panel A. Exploratory factor analysis			
Trust in AI tools	A1 to A4, B1 to B4, In1 to In4	0.75−0.89	0.17
Short-term orientation	I1 to I7	0.82−0.85	0.10
AI-assisted investment decisions	BI1 to BI3, AS1 to AS3	0.75−0.88	0.10
Panel B. Heterotrait–monotrait ratio (HTMT)			
Pair	*Comparison*	*HTMT*	*Above 0.90*
Ability and benevolence	Within trust	1.00	Yes
Ability and integrity	Within trust	1.02	Yes
Benevolence and integrity	Within trust	1.02	Yes
Behavioral intention and actual use	Within investment decisions	1.02	Yes
Short-term orientation and trust	Between constructs	0.46	No
Short-term orientation and investment decisions	Between constructs	0.49	No
Trust and investment decisions	Between constructs	0.49	No

Note: Panel A reports principal-component factors with promax rotation for all 25 items. The first four eigenvalues are 11.20, 3.58, 2.47 and 0.62, and the three retained factors explain 69.0% of the variance. The first unrotated component explains 44.8%. Loadings and cross-loadings are absolute values. Henseler et al. (2015) suggest that HTMT ratios should stay below 0.90.

The first unrotated component accounts for less than half of the total variance, below the level commonly used in Harman's single-factor test. A single common-method factor therefore does not dominate the data, although this test cannot rule out method bias altogether (Podsakoff et al., 2003).

The factor structure does not support the finer distinctions built into the questionnaire. The heterotrait–monotrait ratios (HTMT) among ability, benevolence and integrity are close to one, and so is the ratio between behavioral intention and actual use. All of them exceed the ceiling suggested by Henseler et al. (2015). In other words, respondents did not separate the three trust

dimensions, and they answered the intention items in much the same way as the use items. One likely reason is that many respondents have limited experience with AI investment tools and judge them as a whole rather than facet by facet. The ratios among short-term orientation, trust and investment decisions are well within the acceptable range. Overall trust is therefore the main moderator in what follows. The dimension models are still reported for comparison with earlier ABI studies, but they cannot show which dimension matters most.

4.4 Benchmark results

Table 7. Short-term orientation, trust and AI-assisted investment decisions

	(1) Controls	(2) Trust	(3) Ability	(4) Benevolence	(5) Integrity	(6) All three
Short-term orientation		0.314***	0.331***	0.319***	0.325***	0.330***
		(0.079)	(0.078)	(0.078)	(0.079)	(0.079)
Trust		0.283***				
		(0.081)				
Short-term orientation × Trust		0.106*				
		(0.058)				
Ability			0.247***			0.056
			(0.075)			(0.159)
Short-term orientation × Ability			0.107*			
			(0.056)			
Benevolence				0.270***		0.210
				(0.076)		(0.166)
Short-term orientation × Benevolence				0.106*		
				(0.056)		
Integrity					0.253***	0.049
					(0.079)	(0.179)
Short-term orientation × Integrity					0.101*	
					(0.058)	
Age	0.080	−0.160	−0.175	−0.166	−0.141	−0.147
	(0.189)	(0.166)	(0.167)	(0.165)	(0.167)	(0.168)
Female	−0.005	−0.130	−0.123	−0.115	−0.136	−0.100
	(0.163)	(0.140)	(0.141)	(0.140)	(0.142)	(0.142)
Educational stage	−0.026	0.097	0.095	0.103	0.093	0.089
	(0.151)	(0.130)	(0.131)	(0.130)	(0.132)	(0.132)
Monthly income	−0.083	0.039	0.049	0.040	0.026	0.045
	(0.155)	(0.133)	(0.134)	(0.133)	(0.134)	(0.135)
Married	−0.041	0.060	0.063	0.070	0.065	0.080
	(0.171)	(0.150)	(0.151)	(0.150)	(0.151)	(0.152)
Other marital status	1.018*	0.270	0.262	0.343	0.245	0.386
	(0.537)	(0.470)	(0.474)	(0.467)	(0.474)	(0.475)
Constant	2.982***	2.837***	2.849***	2.819***	2.842***	2.836***
	(0.511)	(0.437)	(0.440)	(0.436)	(0.440)	(0.443)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Equality test, p-value						0.814
Observations	162	162	162	162	162	162
R^2	0.030	0.307	0.297	0.308	0.296	0.294
Adjusted R^2	−0.008	0.266	0.256	0.267	0.255	0.247
F statistic	0.80	7.48***	7.15***	7.53***	7.11***	6.28***

Note: The table reports OLS estimates. The dependent variable is the investment decisions score, the mean of the six intention and use items. Short-term orientation and the trust measures are mean-centered. The reference groups are male and unmarried respondents. Standard errors are in parentheses. In column (2), variance inflation factors for the three core terms are below 1.5. In column (6) they range from 5.6 to 6.9 for the three dimensions, and the equality test is $F(2, 151) = 0.21$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Column (1) of Table 7 regresses investment decisions on the control variables alone. None of the controls is significant at the 5% level, and together they explain very little of the variation in investment decisions. The only coefficient close to significance belongs to the group with other marital status, which has just four members, so it is not interpreted.

Column (2) adds short-term orientation, trust and their interaction. The model as a whole is highly significant and explains close to a third of the variation in investment decisions. With demographic characteristics held constant, short-term orientation has a positive effect that is significant at the 1% level, which supports Hypothesis 1. Respondents with a stronger short-term orientation report more intention to use AI tools and more actual use. Trust also has a positive effect of its own, again significant at the 1% level. The interaction between short-term orientation and trust is positive but significant only at the 10% level, and variance inflation factors show that multicollinearity is not a concern.

Because the sample is small and the t-test assumes normally distributed errors, the interaction coefficient is also bootstrapped. The bootstrap 95% confidence interval excludes zero, though only narrowly (see the note to Table 9). The moderating effect of trust is therefore positive but modest in size.

4.5 Dimensions of trust

Columns (3) to (5) of Table 7 replace overall trust with each of its dimensions, and the results closely follow column (2). In each model, short-term orientation stays positive and significant at the 1% level. The dimension itself also has a positive effect, again significant at the 1% level. The interaction terms are all positive and significant at the 10% level. Perceived ability, benevolence and integrity each appear to strengthen the link between short-term orientation and AI-assisted investment decisions, which offers partial support for Hypotheses 2a, 2b and 2c.

The similarity is expected given the validity results in Section 4.3, since the three scores measure largely the same thing. Column (6) confirms this. When ability, benevolence and integrity enter the same model, none of them is significant, their variance inflation factors rise sharply, and the hypothesis that their coefficients are equal cannot be rejected. The data therefore do not show that one dimension matters more than the others, and Hypotheses 2a to 2c are best read as three versions of the same test.

4.6 How trust changes the effect of short-term orientation

Figure 7 shows how the effect of short-term orientation changes with trust. The effect is positive across the whole observed range of trust and grows steadily as trust rises. The Johnson–Neyman boundary sits well below mean trust, and above it the effect is significant at the 5% level. Most respondents, 138 of the 162, fall in this region. For the small group with very low trust the confidence interval includes zero, so for them it cannot be said that short-term orientation makes any difference to AI use.

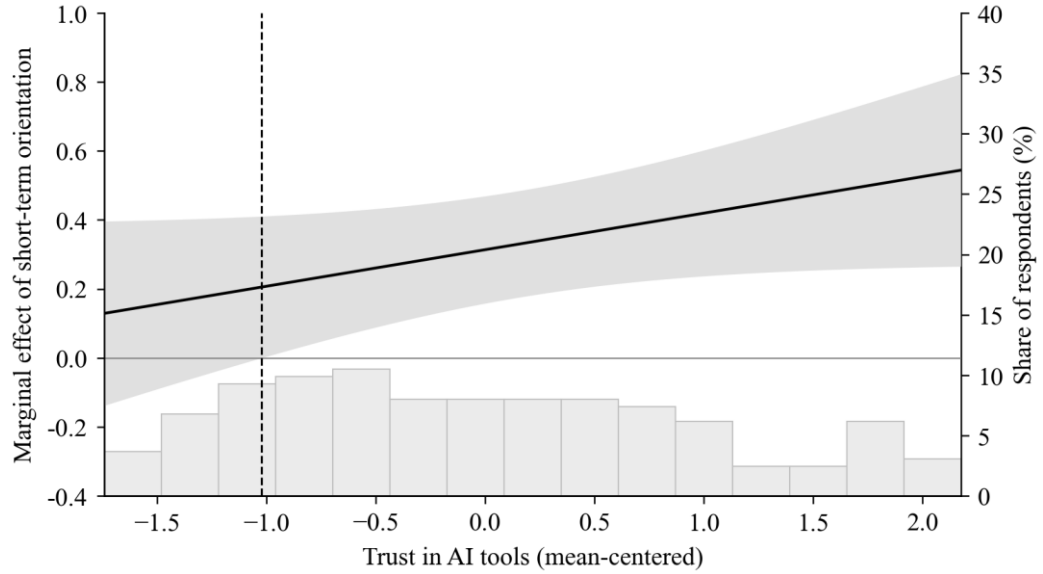
Table 8. Effect of short-term orientation at different levels of trust

Dependent variable	Low trust (−1 SD)	Mean trust	High trust (+1 SD)
Investment decisions	0.210**	0.314***	0.418***
	(0.102)	(0.079)	(0.092)
Behavioral intention	0.206*	0.300***	0.394***
	(0.113)	(0.087)	(0.101)
Actual use	0.214**	0.328***	0.441***
	(0.103)	(0.079)	(0.092)

Note: Each entry is the slope of short-term orientation from a model with trust, the interaction term and the control variables. Trust is mean-centered and one standard deviation equals 0.98. Standard errors are in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

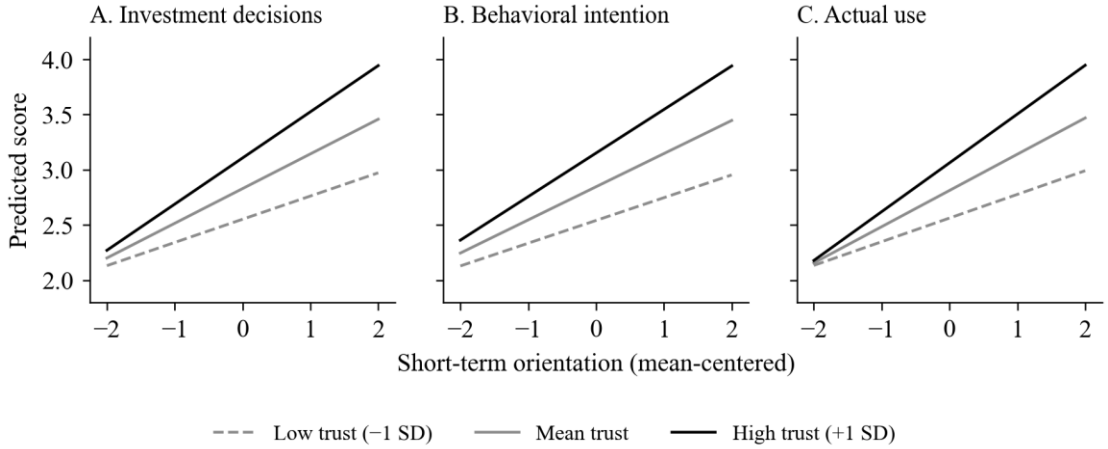
Table 8 reports the simple slopes. At one standard deviation below mean trust, the effect of short-term orientation is significant at the 5% level. At one standard deviation above the mean it is about twice as large and significant at the 1% level. Figure 8 plots the predicted values. The lines fan out to the right. Among respondents with a weak short-term orientation trust makes little difference, whereas among strongly short-term-oriented respondents high trust goes with clearly more AI use.

Figure 7. Marginal effect of short-term orientation across levels of trust



Note: The solid line shows the marginal effect of short-term orientation on investment decisions at each level of trust, based on column (2) of Table 7. The shaded band is the 95% confidence interval. The dashed vertical line marks the Johnson–Neyman boundary, to the right of which the effect is significant at the 5% level. Bars show the distribution of trust (right axis).

Figure 8. Predicted scores at low, mean and high trust



Note: Lines show predicted scores at one standard deviation below mean trust, at mean trust and at one standard deviation above it, averaged over the observed values of the control variables. Confidence intervals are omitted for readability, and Table 8 reports the slopes with their standard errors.

4.7 Robustness

To test whether these patterns hold across both parts of the dependent variable, the score is split into behavioral intention and actual use. Table 9 reports the results together with further checks.

The main effect of short-term orientation holds for both outcomes and is significant at the 1% level in columns (4) and (5). The moderating effect, however, appears in actual use rather than in intention. For actual use the interaction is significant at the 5% level and its bootstrap confidence interval excludes zero. For intention it is smaller and not significant, and the bootstrap interval includes zero. Panels B and C of Figure 8 show the same pattern, with the lines fanning out slightly more for actual use. One reading comes from Mayer et al. (1995), who define trust as a willingness to be vulnerable to another party. Saying that one intends to use a tool costs nothing, whereas letting it shape a real trade puts money at risk, so trust should matter more at that second step. This interpretation is tentative, because the two outcome scales are highly correlated and could not be separated in the validity analysis.

The other checks leave the picture unchanged. HC3 robust standard errors leave the significance levels nearly unchanged, and dropping item I3 from the short-term orientation scale or AS3 from the use scale changes the coefficients only slightly. The interaction stays positive in every specification.

Table 9. Robustness checks

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Short-term orientation	0.314***	0.314***	0.317***	0.300***	0.328***	0.328***	0.316***
	(0.079)	(0.074)	(0.077)	(0.087)	(0.079)	(0.074)	(0.087)
Trust	0.283***	0.283***	0.288***	0.311***	0.255***	0.255***	0.281***
	(0.081)	(0.091)	(0.080)	(0.089)	(0.081)	(0.090)	(0.089)
Short-term orientation × Trust	0.106*	0.106*	0.101*	0.096	0.116**	0.116**	0.129**
	(0.058)	(0.056)	(0.057)	(0.064)	(0.058)	(0.055)	(0.064)
Dependent variable	Decisions	Decisions	Decisions	Intention	Use	Use	Use
Short-term orientation items	7	7	6	7	7	7	7
Actual use items	3	3	3		3	3	2
Standard errors	OLS	HC3	OLS	OLS	OLS	HC3	OLS
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	162	162	162	162	162	162	162
R^2	0.307	0.307	0.309	0.265	0.312	0.312	0.293

Note: Column (1) repeats column (2) of Table 7. Columns (2) and (6) use HC3 robust standard errors. Column (3) drops item I3 from the short-term orientation scale, and column (7) drops item AS3 from the actual use scale. All models include the control variables. Standard errors are in parentheses. Bootstrap 95% percentile intervals for the interaction, from 2,000 resamples, are [0.004, 0.222] in column (1), [−0.013, 0.222] in column (4) and [0.011, 0.235] in column (5). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5. Conclusion

This study used 162 questionnaires from Macau to examine how short-term orientation relates to AI-assisted investment decisions and whether trust moderates that relationship.

The effect of short-term orientation on AI-assisted investment decisions is positive and significant in every specification, which supports Hypothesis 1. Respondents with a stronger short-term orientation report more intention to use AI tools for investing and more actual use. The finding is consistent with time discounting theory (Loewenstein & Prelec, 1992) and scarcity theory (Mullainathan & Shafir, 2013). The survey did not measure economic stress. The claim that stress shortens time horizons and pushes investors toward AI is therefore an interpretation, not a tested pathway.

Trust in AI has a positive effect of its own and strengthens the link between short-term orientation and AI use. The interaction is modest. It is only marginally significant in the conventional test, but its bootstrap confidence interval excludes zero, so Hypotheses 2a to 2c receive partial support. For high-trust respondents the effect of short-term orientation is about twice that for low-trust respondents. The moderation is clearer for actual use than for intention, which fits the view of trust as a willingness to accept risk (Mayer et al., 1995).

For AI platform designers, the results suggest that trust works as a whole. Respondents did not judge ability, benevolence and integrity separately, so improving one of them alone is unlikely to be enough. Platforms should explain clearly how their recommendations serve the user's financial interests rather than anyone else's, and avoid black-box advice. They should also make their tools easier to use so that trust carries through to the point of action. Trust, however, also amplifies short-term behavior. A platform that wins its users' trust without paying attention to how they use it could end up speeding up the decisions of its most short-term-oriented users. Features that show the long-run consequences of a trade, or ask for confirmation before frequent trading, could keep that trust from working against users.

For financial regulators, the concern is what happens if economic stress pushes many investors toward a short-term orientation at the same time. Regulatory frameworks would then need to anticipate stronger herding and volatility. This is also where the study connects to sustainable financial development. When many investors rely on similar tools at the same moment, their trades become correlated, which may add to volatility. Two measures would help keep the

spread of AI tools in line with long-term financial stability. Regulators could require robo-advisers to disclose how they produce recommendations, and investor education could explain present bias.

6. Limitations and future research

The study has several limitations. The data are cross-sectional and self-reported, so the associations cannot be read as causal. Investors who already use AI tools may, for instance, come to think in shorter terms. The snowball sample is small and skewed toward well-educated, middle-income adults in Macau, which limits how far the results generalize. Economic stress, the starting point of the argument, was not measured either. A future survey could add a financial-stress scale and test the full chain from stress to short-term orientation to AI use.

The measures also need work. Item I3 behaved like a forward-worded item, the three trust dimensions could not be told apart, and the intention and use scales were almost identical in the data. An established scale for trust in automated systems (Jian et al., 2000) would give a firmer measure of overall trust, and an experiment that varies the information people receive about a tool's benevolence would allow a sharper test of the ABI dimensions. Actual use was also self-reported over three months. Platform records or a panel design would give a more reliable measure of behavior.

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